Q&A
THINGS YOU WANTED TO KNOW!

Ques. HOW CAN I DESIGN FERRITE TRANSFORMERS OF VARIOUS TOPOLOGIES? PLEASE SUGGEST SOME ONLINE STORES WHERE FERRITE TRANSFORMER COMPONENTS (CORE, BOBBINS, ETC) ARE AVAILABLE ON LOW VOLUME.

A. Samiuddhin

Ans. A ferrite transformer has a magnetic core in which coil (inductor) windings are made on a ferrite core component. It offers low eddy current losses. It is normally used for high-frequency applications. Common ferrite core types are toroidal, closed-core, shell and cylindrical.

Depending on circuit designs, core types and applications of transformers, there are different topologies and names. These include shell type, push-pull, half-bridge and flyback. Irrespective of the topologies, some points to be kept in mind while designing ferrite transformers include frequency and temperature of operation, unit cost, size and shape. These should match the voltage levels of source and load, provide electrical isolation, prevent core saturation and minimise core losses.

The size and frequency of operation of a ferrite transformer depends on two broad applications: signal and power. A ferrite transformer used in signal applications is small and has higher frequencies (in the range of mega-Hertz). The one used in power applications is large and has lower frequencies (normally ranging from 1kHz to 200kHz).

Steps for designing a ferrite transformer

Application. Before designing a transformer, check your requirement and exact application. This may include input voltage, output voltage, current and frequency of operation. Then, consider other parameters like physical size, spacing, mounting style, isolation, leakage currents and temperature.

Core selection. Most core types need bobbins to fit the cores you choose, and assist in the mounting of the finished product. Make sure the bobbin style and materials are available in the local market. Then, calculate the correct number of turns, power losses and other parameters. You may refer to the formulae given on the following websites:

- *www.electronicdesign.com/power/build-your-own-transformer,*
- *www.electrical4u.com/design-of-high-frequency-pulse-transformer/*
- *http://ecee.colorado.edu/~ecen5797/course_material/Ch15slides.pdf*

Winding. Primary winding current and wire size need to be determined. Primary current is equal to total output power plus transformer power losses, divided by primary voltage.


Fig. 1: Different types of ferrite cores (Credit: www.yeng-tat.com)


Fig. 2: Different types of bobbins (Credit: www.ramsales.net)

Next comes the number of turns needed for secondary winding. For this, check if the wires fit into your winding area in the bobbin, height and mean length of turns from the mechanical drawing. Include insulation used between windings, while considering total winding height.

Verification. Verify the design by measuring open-circuit and loaded voltages across secondary winding. For this, calculate resistance of each winding. Then, calculate voltage drop across that winding by multiplying resistance and current in the winding.

Calculate open-circuit voltage and loaded voltage across secondary winding using the formulae given in the above-mentioned websites.

Temperature calculation. Acceptable temperature rise depends on the application and the designer. Two main causes of temperature rise in a transformer are core power losses and winding power losses. These can be calculated using standard formulae.

You can buy ferrite transformers from online stores like *www.indiamart.com*, *www.rrcomponent.in* and *www.digikey.com*

Q2. I HAVE TWO MAGNETIC COILS FACING EACH OTHER. WHEN POWER IS APPLIED TO THE FIRST COIL, WILL ITS MAGNET FIELD CREATE A MAGNET FIELD IN THE SECOND? CAN THE MAGNETIC FIELD IN THE SECOND COIL TURN ON THE ELECTRICAL LOAD CONNECTED TO IT?

Richard Shambley

A2. Yes! According to the property of mutual induction, any change in current in one coil induces an electromotive force, or emf, in another coil. The induced emf is proportional to the number of turns on

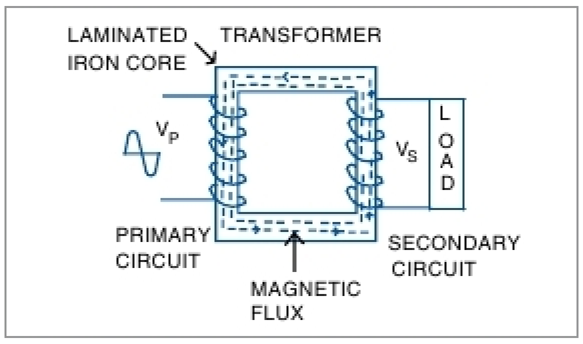

Fig. 3: Induced electromotive force in transformer

the secondary coil and change in current in the first coil. If the power in the first coil is switched on and off, emf will be induced in the second coil.

Current is the result of an emf induced by a changing magnetic field. If there is a complete circuit, induced current will flow through the circuit, including load.

Iron has high magnetic permeability and, hence, is easily magnetised. If two coils are placed on the same iron core, feed AC supply (changing current) to the first (primary) coil. Emf will generate on the other (secondary) coil due to mutual induction. This principle is used in a transformer, too.

In your case, when you feed AC power supply to the first coil, you can easily operate the load connected across the second coil.

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